

WASP-12b

R-1740

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_190322 · created 2026-08-02T18:28:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458564.69108 +/- 0.00059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1 +/- 0.039
Transit depth (Rp/Rs)^2	1.0 +/- 0.78 [%]
Orbital Inclination (inc)	78.84 +/- 2.8
Airmass coefficient 1 (a1)	0.942 +/- 0.015
Airmass coefficient 2 (a2)	0.0334 +/- 0.0082
Scatter in the residuals of the lightcurve fit is	1.62 %
Best Comparison Star	#1 - [446, 265]
Optimal Aperture	3.65
Optimal Annulus	7.91
Transit Duration (day)	0.0849 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1 $\pm$ 0.039 vs 0.1178 (deviation 0.46 σ)
Duration fit vs published	0.0849 d vs 0.125 d (deviation 6.08 σ)
Mid-time O-C	-1.9 min
Depth SNR	2.6
Residual scatter	1.62 %

Light curve

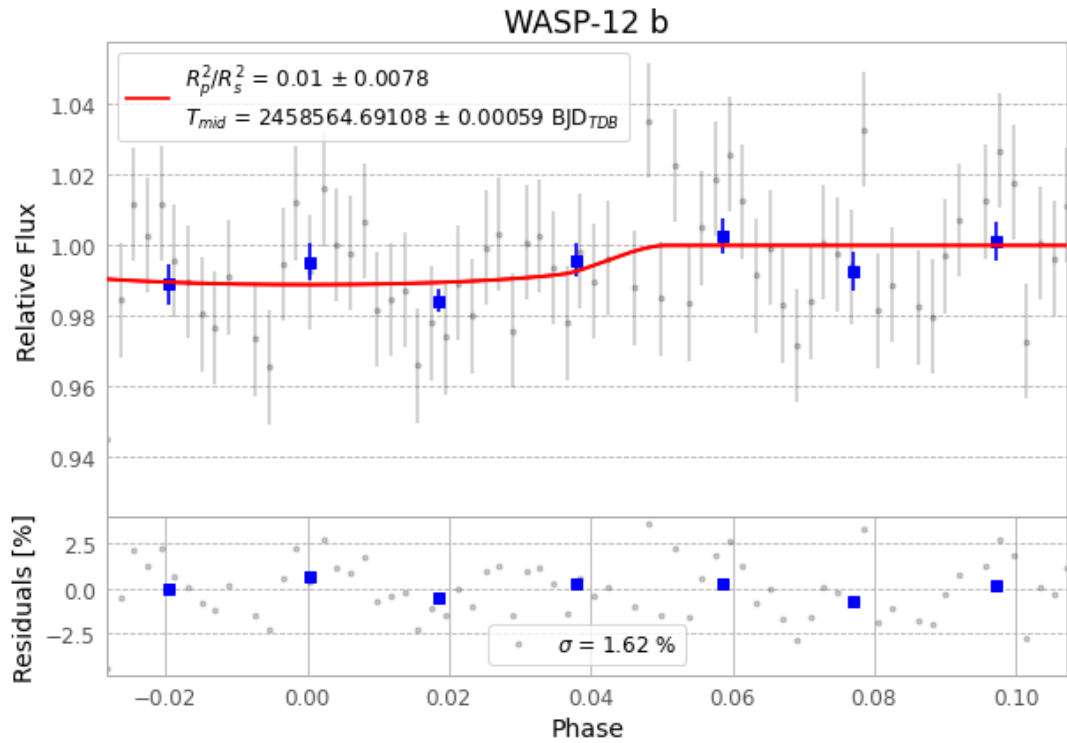


Figure 1 — FinalLightCurve_WASP-12 b_2019-03-21.png (SHA-256 7a46587bd8bed1d7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 606ace5b867fe3db...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), WASP-12190322034812.FITS → WASP-12190322072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-190322.log (9d14220adef5...), FinalLightCurve_WASP-12 b_2019-03-21.png (7a46587bd8be...), FinalLightCurve_WASP-12 b_2019-03-21.csv (6fa46adc5879...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 18:28Z.